

What is the evidence for psilocybin in treatment-resistant depression?

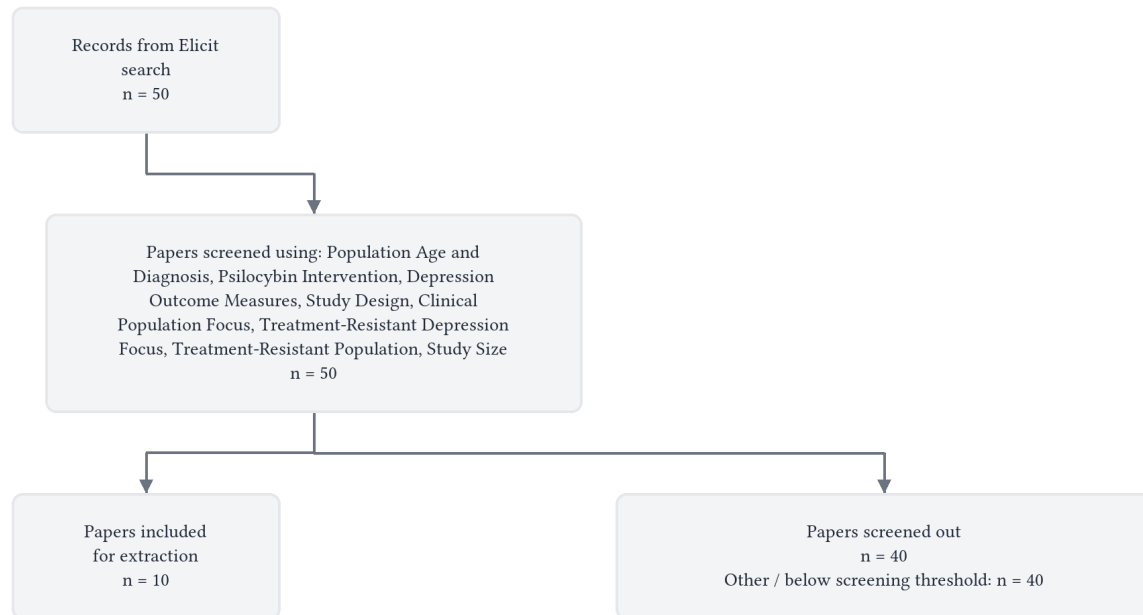
Evidence from multiple trials including the largest randomized controlled trial demonstrates that psilocybin produces rapid, robust antidepressant effects in treatment-resistant depression, though sustained response is limited and serious adverse events including suicidal ideation occurred in a minority of patients.

Abstract

Psilocybin demonstrates robust short-term antidepressant effects in treatment-resistant depression across 10 studies comprising 5 open-label trials, 2 randomized controlled trials, and 3 secondary analyses. In the largest randomized trial (n=233), a single 25 mg dose reduced depression scores significantly more than 1 mg control at 3 weeks (MADRS difference -6.6, 95% CI -10.2 to -2.9, $P < 0.001$) [1], with response rates of 37% versus 18% [1]. Open-label studies reported even larger effect sizes ranging from Hedges' $g = 1.27$ to 3.1 at 1-3 weeks [2, 3], though these likely reflect placebo and expectancy effects alongside genuine treatment response. Depression reductions occurred within 1 week [2, 4, 5] and remained statistically significant at 6 months in follow-up studies (Cohen's $d = 1.4$ - 1.5) [4, 5], but sustained response rates at 12 weeks were only 20% in the randomized trial [1], indicating substantial relapse over time. The quality of the acute psychedelic experience predicted clinical outcomes [2-4, 6], while comorbid PTSD was associated with reduced treatment response [6].

Safety profiles were generally favorable in smaller studies with no serious adverse events [2-4, 7], but the largest trial identified serious adverse events including suicidal ideation, intentional self-injury, and psychiatric hospitalization [1], with worsening suicidal state in 14% of the 25 mg group [1]. Transient anxiety during dosing was universal [2, 4], and common adverse events included headache (24%) and nausea (22%) [1]. The evidence base consists primarily of patients with unipolar depression [1, 2, 4-6, 8] who had failed 2-5 previous treatments [1, 2, 4, 6], with systematic exclusion of psychotic disorders [2, 4, 5], active substance use [2, 9], and high suicide risk in most studies [1]. Psilocybin shows promise for treatment-resistant depression based on convergent short-term efficacy findings, but durability of response, optimal dosing strategies, and safety in broader clinical populations require further investigation in larger controlled trials with extended follow-up and active comparators.

Flow Diagram



Paper search

We performed a semantic search across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We ran this query: "What is the evidence for psilocybin in treatment-resistant depression?"

The search returned 50 total results from Elicit.

We retrieved 50 papers most relevant to the query for screening.

Screening

We screened in sources based on their abstracts that met these criteria:

- **Population Age and Diagnosis:** Does the study include adults (≥ 18 years) diagnosed with treatment-resistant depression defined as failure to respond to at least two adequate trials of antidepressant medications?
- **Psilocybin Intervention:** Does the intervention involve psilocybin administration in any form (pure psilocybin, psilocybin-containing mushrooms, or synthetic psilocybin)?
- **Depression Outcome Measures:** Does the study measure depression-related outcomes using validated scales (e.g., MADRS, HAM-D, BDI, PHQ-9)?
- **Study Design:** Is the study design a randomized controlled trial, controlled clinical trial, cohort study, case-control study, systematic review, or meta-analysis?

- **Clinical Population Focus:** Does the study focus on a clinical population rather than solely on healthy volunteers or recreational use?
- **Treatment-Resistant Depression Focus:** Is psilocybin used primarily for treatment-resistant depression (rather than primarily for other conditions like anxiety, PTSD, or addiction without specific focus on treatment-resistant depression)?
- **Treatment-Resistant Population:** Do the participants have depression that is characterized as treatment-resistant (rather than treatment-naïve depression)?
- **Study Size:** If the study is a case report or case series, does it include 5 or more participants?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

At abstract screening, the number of papers excluded for each primary reason was:

- **Other / below screening threshold:** n = 40

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Study Design:**

Extract study design and methodological details relevant to assessing evidence quality for psilocybin in treatment-resistant depression, including:

- Study design type (RCT, open-label, pilot, etc.)
- Sample size and randomization details
- Study duration and follow-up periods
- Control/comparison conditions (if any)
- Setting and location

- **TRD Population:**

Extract how treatment-resistant depression was defined and characterized in this study, including:

- Specific criteria for treatment resistance (number of failed treatments, types of treatments)
- Depression severity at baseline (MADRS, HAM-D, etc. scores)
- Primary diagnosis (unipolar vs bipolar depression)
- Key exclusion criteria and comorbidities (especially PTSD, substance use, psychosis)
- Participant demographics and baseline characteristics

- **Psilocybin Protocol:**

Extract all details about the psilocybin intervention protocol for treating treatment-resistant depression, including:

- Psilocybin dose(s) and formulation (synthetic vs natural)
- Number and timing of dosing sessions
- Psychological support structure (preparation, integration sessions)
- Concomitant medications (discontinuation requirements, ongoing SSRIs)
- Setting and supervision during dosing sessions

- **Depression Outcomes:**

Extract all depression-related efficacy outcomes and results for psilocybin treatment of treatment-resistant depression, including:

- Primary and secondary depression measures used (MADRS, QIDS, PHQ-9, etc.)
- Baseline and follow-up scores at all timepoints
- Effect sizes (Cohen's d, Hedges' g) and confidence intervals
- Response and remission rates with definitions used
- Statistical significance and p-values
- Time to response and durability of effects

- **Safety Profile:**

Extract safety and tolerability data for psilocybin treatment in treatment-resistant depression patients, including:

- All adverse events (mild, moderate, severe) with frequencies
- Serious adverse events and hospitalizations
- Suicidal ideation or behavior changes
- Study discontinuations due to adverse events
- Acute psychological distress during dosing sessions
- Any safety concerns specific to the TRD population

- **Response Predictors:**

Extract factors that predicted or moderated treatment response to psilocybin in treatment-resistant depression, including:

- Baseline characteristics associated with better/worse outcomes
- Psychedelic experience factors (mystical experiences, oceanic boundlessness, etc.)
- Individual variability in response patterns
- Subgroup analyses (by comorbidity, treatment resistance level, etc.)
- Factors affecting durability of response
- Any mechanistic insights about who benefits most

Results

Characteristics of Included Studies

The systematic review included 10 studies examining psilocybin for treatment-resistant depression, comprising 5 open-label trials, 2 randomized controlled trials, 2 secondary analyses, and 1 pilot trial. Sample sizes ranged from 7 to 233 participants. Two studies involved neuroimaging analyses of previously conducted trials.

Study	Full text retrieved?	Study Design	Sample Size	TRD Criteria	Primary Diagnosis	Follow-up Duration
R. Carhart-Harris et al., 2016 [2]	Yes	Open-label feasibility trial [2]	12 [2]	≥2 failed treatments, ≥6 weeks each [2]	Unipolar depression [2]	3 months [2]
G. Goodwin et al., 2022 [1]	Yes	Phase 2 double-blind RCT [1]	233 [1]	2-4 failed treatments, ≥8 weeks each [1]	Major depressive disorder without psychotic features [1]	12 weeks [1]
R. Carhart-Harris et al., 2017 [4]	Yes	Open-label feasibility study [4]	20 [4]	≥2 failed treatments, ≥6 weeks each [4]	Unipolar depression [4]	6 months [4]
G. Goodwin et al., 2023 [7]	Yes	Phase II open-label study [7]	19 [7]	2-4 failed treatments for current episode [7]	Unipolar depression (MDD without psychotic features) [7]	3 weeks [7]
R. Carhart-Harris et al., 2017a [8]	Yes	Open-label clinical trial with fMRI [8]	19 (16 analyzed) [8]	Not explicitly stated [8]	Unipolar depression [8]	5 weeks [8]
J. Rosenblatt et al., 2024 [9]	No	Randomized clinical trial [9]	30 [9]	High levels of treatment resistance in MDD or bipolar II disorder [9]	Major depressive or bipolar II disorder [9]	6 months [9]
G. Goodwin et al., 2023a [10]	No	Phase 2 RCT (secondary analysis) [10]	233 [10]	Not specified [10]	Not specified [10]	3 weeks [10]
S. Aaronson et al., 2025 [6]	No	Open-label trial [6]	12 [6]	≥5 failed treatments [6]	Unipolar depression (implied) [6]	12 weeks [6]
Sally Meikle et al., 2025 [3]	No	Open-label, single-arm pilot trial [3]	7 [3]	Not specified [3]	Not specified [3]	20 weeks [3]
Richard E. Daws et al., 2022 [5]	Yes	Open-label trial and phase II RCT with fMRI [5]	19 (open-label), 59 (RCT) [5]	Mean 4.6 failed medications [5]	Unipolar depression [5]	6 months (open-label), 3 weeks (RCT) [5]

The studies enrolled participants with moderate to severe depression at baseline. In studies reporting MADRS scores, baseline means ranged from 31.7 to 33 [1, 7]. Studies using the QIDS-SR16 reported baseline scores of 18.9 [8], while one study using the Beck Depression Inventory reported a mean baseline score of 34.81 [5]. Most participants had failed multiple previous treatments, with one study requiring failure of at least five treatments for inclusion [6]. Depression severity was typically classified as severe in 57.9% to 100% of participants [2, 7]. Key exclusion criteria across studies included current or past psychotic disorders [2, 4, 5], while some studies excluded those with substance use disorders [2, 9] or at clinically significant risk for suicide [1].

Psilocybin Treatment Protocols

Psilocybin doses and administration varied across studies. The most common protocol involved two dosing sessions of 10 mg and 25 mg administered 7 days apart [2–4]. The largest randomized trial compared single doses of 25 mg, 10 mg, and 1 mg (control) [1]. One study specifically examined severe TRD using a single 25 mg dose [6], while another investigated one, two, or three 25 mg sessions over 6 months [9].

Study	Psilocybin Formulation	Dose(s)	Number of Sessions	Psychological Support	Concomitant Medications
R. Carhart-Harris et al., 2016 [2]	Synthetic [2]	10 mg, 25 mg [2]	2 sessions, 7 days apart [2]	4-hour preparation, support during and after sessions [2]	May need to withdraw antidepressants [2]
G. Goodwin et al., 2022 [1]	Synthetic (COMP360) [1]	25 mg, 10 mg, or 1 mg (control) [1]	Single administration [1]	3 preparation sessions, 2 integration sessions [1]	Discontinued ≥ 2 weeks before dosing, off for 3 weeks post-dosing [1]
R. Carhart-Harris et al., 2017 [4]	Synthetic (capsules) [4]	10 mg, 25 mg [4]	2 sessions, 7 days apart [4]	Preparation, acute support, integration [4]	Discontinued ≥ 2 weeks before trial (one exception) [4]
G. Goodwin et al., 2023 [7]	Synthetic (COMP360) [7]	25 mg [7]	Single administration [7]	3 preparation sessions, integration sessions at Day 2 and Week 1 [7]	Continued on prescribed SSRI throughout study [7]
R. Carhart-Harris et al., 2017a [8]	Not specified [8]	10 mg, 25 mg [8]	2 sessions, 1 week apart [8]	Not mentioned [8]	Not mentioned [8]
J. Rosenblat et al., 2024 [9]	Not mentioned [9]	25 mg [9]	1, 2, or 3 sessions [9]	Preparation and integration psychotherapy sessions [9]	Not mentioned [9]

Study	Psilocybin Formulation	Dose(s)	Number of Sessions	Psychological Support	Concomitant Medications
G. Goodwin et al., 2023a [10]	Synthetic [10]	25 mg, 10 mg, or 1 mg (control) [10]	Single dose [10]	Psychological support from trained therapists [10]	Not mentioned [10]
S. Aaronson et al., 2025 [6]	Synthetic [6]	25 mg [6]	Single dose during 8-hour dosing day [6]	3 pre-treatment sessions, support during dosing, 3 post-treatment integration sessions [6]	Discontinued ≥ 2 weeks before dosing, off for ≥ 3 weeks post-dosing [6]
Sally Meikle et al., 2025 [3]	Not mentioned [3]	25 mg [3]	2 sessions [3]	3 preparatory sessions, 6 integration sessions [3]	Not mentioned [3]
Richard E. Daws et al., 2022 [5]	Synthetic [5]	Open-label: 10 mg, 25 mg; RCT: 25 mg or 1 mg [5]	Open-label: 2 sessions, 7 days apart; RCT: 2 sessions, 3 weeks apart [5]	Not explicitly mentioned [5]	Exclusion of SSRI contraindications in RCT [5]

All studies providing details employed psychological support structures including preparation sessions before dosing and integration sessions afterward. Dosing sessions lasted 6-8 hours [1, 6] in supervised settings with trained therapists present [1, 7]. Most studies required discontinuation of concomitant antidepressants at least 2 weeks before dosing [1, 4, 6], with one notable exception examining psilocybin adjunct to ongoing SSRI treatment [7].

Effects on Depression Severity

Short-term Outcomes (1-3 Weeks)

Studies consistently demonstrated rapid reductions in depression severity. In the first open-label trial, mean QIDS scores decreased by 11.8 points at 1 week post-treatment (95% CI -9.15 to -14.35, $p=0.002$, Hedges' $g=3.1$) [2]. The largest randomized controlled trial found that 25 mg psilocybin reduced MADRS scores by 12.0 points at week 3, compared to 5.4 points with 1 mg control (difference -6.6, 95% CI -10.2 to -2.9, $P<0.001$) [1]. The 10 mg dose showed intermediate effects with a reduction of 7.9 points, though this did not differ significantly from the 1 mg control (difference -2.5, 95% CI -6.2 to 1.2, $P=0.18$) [1].

In the study examining psilocybin adjunct to SSRIs, participants experienced a mean MADRS reduction of 14.9 points at week 3 (95% CI -20.7 to -9.2) [7]. The severe TRD study reported MADRS least-squares mean changes of -15.8 at week 3 (95% CI -25.4 to -6.3) [6]. A pilot trial found a mean QIDS change of -7.14 at 3 weeks ($p=0.02$, Hedges' $g=-1.27$, 95% CI -2.40 to -0.37) [3].

Neuroimaging studies corroborated clinical findings. One study using the Beck Depression Inventory reported a mean reduction of 21.0 points at 1 week (Cohen's $d=1.78$, $P<0.001$) [5]. Another found QIDS-SR16 scores decreased from baseline 18.9 ± 3 to 8.8 ± 6.2 at 1 week post-treatment ($p<0.001$) [8].

Medium-term Outcomes (5 Weeks to 3 Months)

Clinical improvements remained substantial at intermediate timepoints. At 5 weeks, the extended follow-up study reported Cohen's $d=2.3$ for QIDS-SR16 reductions ($p<0.001$) [4], with 47% of patients meeting response criteria ($\geq 50\%$ reduction) [8]. At 3 months, the original feasibility study showed mean QIDS reductions of 9.2 points (95% CI -5.69 to -12.71, $p=0.003$, Hedges' $g=2$) [2], with 58% maintaining response [2].

Long-term Outcomes (6 Months to 1 Year)

Longer-term follow-up revealed persistent but attenuating effects. At 6 months, effect sizes remained significant with Cohen's $d=1.4$ ($p<0.001$) for QIDS-SR16 [4] and Cohen's $d=1.07$ ($P<0.001$) for BDI [5]. The severe TRD study reported maintained improvements at 12 weeks with MADRS least-squares mean change of -17.2 (95% CI -25.2 to -9.1) [6]. However, the largest RCT found sustained response rates at 12 weeks were only 20% for 25 mg, 5% for 10 mg, and 10% for 1 mg control [1], indicating substantial relapse over time.

Response and Remission Rates

Response and remission rates varied across studies and timepoints. At 3 weeks in the largest RCT, response rates ($\geq 50\%$ MADRS reduction) were 37% for 25 mg, 19% for 10 mg, and 18% for 1 mg [1]. Remission rates (MADRS ≤ 10) were 29% for 25 mg, 9% for 10 mg, and 8% for 1 mg [1]. The SSRI adjunct study found both response and remission in 42.1% of participants at week 3 [7].

Earlier open-label studies reported higher rates: 67% achieved remission at 1 week in the initial feasibility trial [2], while the 6-month follow-up study found 9 of 20 patients (45%) met response criteria at week 5, with 4 achieving remission (20%) [4].

The pilot trial revealed marked individual variability: two participants showed sustained response, three relapsed, and two exhibited no substantial improvement [3]. This heterogeneity in outcomes underscores the need for understanding response predictors.

Safety and Tolerability

The safety profile of psilocybin in treatment-resistant depression populations was generally favorable across studies, though certain adverse events occurred consistently.

Study	Common Adverse Events	Serious Adverse Events	Suicidal Ideation/Behavior	Discontinuations
R. Carhart-Harris et al., 2016 [2]	Transient anxiety (100%), confusion/thought disorder (75%), nausea (33%), headache (33%) [2]	None [2]	Not mentioned [2]	None [2]

Study	Common Adverse Events	Serious Adverse Events	Suicidal Ideation/Behavior	Discontinuations
G. Goodwin et al., 2022 [1]	Headache (24%), nausea (22%), dizziness (6%), fatigue (6%); overall 77% had any AE [1]	Suicidal ideation, intentional self-injury, hospitalization for severe depression [1]	Worsening in 14% (25 mg), 17% (10 mg), 9% (1 mg) [1]	6% (25 mg), 12% (10 mg), 13% (1 mg) [1]
R. Carhart-Harris et al., 2017 [4]	Transient anxiety (n=15), headaches (n=8), nausea (n=5), paranoia (n=3) [4]	None [4]	Significantly reduced at 1 and 2 weeks post-treatment [4]	One patient chose not to complete follow-up [4]
G. Goodwin et al., 2023 [7]	TEAEs in 63.2%; most mild and resolved on day of onset [7]	None [7]	No increase during study [7]	None [7]
R. Carhart-Harris et al., 2017a [8]	Not mentioned [8]	Not mentioned [8]	Not mentioned [8]	Not mentioned [8]
J. Rosenblat et al., 2024 [9]	Transient [9]	None [9]	Not mentioned [9]	Not mentioned [9]
G. Goodwin et al., 2023a [10]	Not mentioned [10]	Not mentioned [10]	Not mentioned [10]	Not mentioned [10]
S. Aaronson et al., 2025 [6]	Not mentioned [6]	Not mentioned [6]	Not mentioned [6]	Not mentioned [6]
Sally Meikle et al., 2025 [3]	Consistent with previous studies [3]	None [3]	Not mentioned [3]	Not mentioned [3]
Richard E. Daws et al., 2022 [5]	Not mentioned [5]	Not mentioned [5]	Not mentioned [5]	Not mentioned [5]

Transient anxiety during drug onset was universal in studies reporting this outcome [2, 4]. Headache and nausea were the most frequently reported treatment-emergent adverse events, affecting 22-33% of participants [1, 2]. These events were typically mild and resolved within hours to days [2, 7]. Confusion or thought disorder occurred in 75% of participants in one study but was transient [2]. No patients required tranquilizing medications during sessions in the initial feasibility trial [2].

The largest randomized trial identified serious adverse events including suicidal ideation, intentional self-injury, and hospitalization for severe depression [1]. Worsening of suicidal state occurred in 14% of the 25 mg group, 17% of the 10 mg group, and 9% of the 1 mg control group [1]. In contrast, another study found suicidal ideation was significantly reduced at 1 and 2 weeks post-treatment [4], while the SSRI adjunct study reported no increase in suicidal ideation during the study [7]. Study discontinuation rates due to adverse events ranged from 6% to 13% in the largest trial [1], while most other studies reported no discontinuations [2, 7].

Factors Associated with Treatment Response

Several factors emerged as potential predictors or moderators of treatment response, though findings varied across studies.

Psychedelic Experience Characteristics: The quality and intensity of the acute psychedelic experience correlated with clinical outcomes in multiple studies. One study found that reductions in depressive symptoms at 5 weeks were predicted by the quality of the acute psychedelic experience [2], with significant relationships between the "USB" factor (experience of unity, spiritual experience, blissful state) and symptom changes [4]. The Oceanic Boundlessness dimension of the psychedelic experience correlated with post-dosing antidepressant responses in severe TRD [6]. A pilot trial identified mindset prior to dosing, spiritual experiences, and perceptual shifts during dosing as predictors of treatment trajectory, while treatment expectations were not a reliable predictor [3]. The intensity of altered or mystical states of consciousness was noted as correlating with therapeutic outcomes [7].

Neural Mechanisms: Neuroimaging studies identified brain connectivity changes associated with treatment response. Increased ventromedial prefrontal cortex-bilateral inferior lateral parietal cortex resting-state functional connectivity predicted response at 5 weeks, as did decreased parahippocampal-prefrontal cortex connectivity [8]. Decreased brain network modularity was associated with improved clinical outcomes and predicted sustained improvements in depression severity [5], with early-phase modularity changes being predictive of long-term response [5]. Changes in network flexibility, particularly in the executive and salience networks, were identified as key components of the therapeutic mechanism [5].

Comorbidity Effects: Patients with comorbid PTSD (n=5) showed significantly less antidepressant effect of psilocybin [6], suggesting this comorbidity may moderate treatment response. Studies generally excluded individuals with psychotic disorders [2, 4, 5] and some excluded substance use disorders [2, 9], limiting knowledge about psilocybin's effects in these populations.

Dose Effects: Pharmacokinetic research showed dose-dependent increases in receptor occupancy and subjective effects [1], with the 25 mg dose demonstrating superior efficacy to 10 mg and 1 mg in the largest trial [1]. Repeated doses were associated with further reductions in MADRS scores compared to baseline [9], though specific dose-response relationships require further investigation.

Individual Variability: One pilot study documented diverse response patterns, with two participants showing sustained response, three experiencing relapse, and two exhibiting no substantial improvement [3]. This heterogeneity suggests important individual differences in treatment response that warrant further study.

Synthesis

The evidence base for psilocybin in treatment-resistant depression reveals consistently robust short-term effects across diverse study designs and populations, but substantial heterogeneity in sustained response rates and concerning safety signals in the largest randomized trial merit careful interpretation.

Short-term Efficacy: Convergent Evidence Across Study Designs

Effect sizes for depression reduction at 1-3 weeks were remarkably consistent across methodologically diverse studies. Open-label trials reported effect sizes ranging from Hedges' $g=1.27$ to 3.1 [2, 3], while the phase 2 RCT demonstrated a statistically significant 6.6-point greater MADRS reduction with 25 mg versus 1 mg control ($P<0.001$) [1]. Even accounting for potential placebo effects in open-label designs and the active control effect of 1 mg psilocybin, the convergence of findings across blinded and unblinded studies supports genuine short-term antidepressant efficacy. The response occurred within 1 week in most studies [2, 4, 5], distinguishing psilocybin from conventional antidepressants which typically require weeks to months for therapeutic effects.

Durability: Attenuation Over Time

The apparent contradiction between open-label studies showing sustained effects at 6 months (Cohen's $d=1.4-1.5$) [4,

5] and the RCT showing only 20% sustained response at 12 weeks [1] may reflect several factors. First, open-label designs likely capture additional placebo effects and expectancy biases that inflate durability estimates. Second, the RCT's stringent definition of sustained response (meeting criteria at week 3 and all subsequent visits) may underestimate partial or fluctuating responses that still represent clinical improvement. Third, sample enrichment may have occurred in open-label studies through selective reporting or participant retention, while the RCT's systematic follow-up captured all patterns including non-responders and relapsers.

The trajectory data suggest initial benefit followed by gradual relapse for many patients. Effect sizes decreased from 2.3 at week 5 to 1.5 at 3 months and 1.4 at 6 months in one study [4], indicating statistical significance was maintained but clinical magnitude diminished. This pattern suggests psilocybin may provide a "window of opportunity" for neuroplastic change rather than indefinite symptom suppression, aligning with proposed "reset" mechanisms [8] and neural network integration changes [5]. Repeated dosing strategies showed promise for enhancing durability [9], though optimal timing and frequency remain undefined.

Dose-Response and Protocol Optimization

The 25 mg dose demonstrated clear superiority to 1 mg control [1], while 10 mg showed intermediate effects that did not reach statistical significance versus control [1]. This suggests a dose threshold for reliable therapeutic effects between 10-25 mg in single-dose paradigms. However, the two-dose protocol (10 mg followed by 25 mg seven days later) used in multiple successful studies [2-4] may provide advantages over single dosing through gradual dose escalation and repeated psychedelic experiences. The feasibility of repeated dosing over 6 months [9, 9] and preliminary evidence for cumulative benefits [9] warrant investigation of maintenance dosing strategies.

The demonstration that psilocybin can be administered safely adjunct to ongoing SSRI treatment [7, 7] has important clinical implications, as antidepressant withdrawal may be problematic or contraindicated for many TRD patients. While SSRIs theoretically attenuate 5-HT_{2A} receptor activation, therapeutic effects were observed in this paradigm [7], suggesting either incomplete receptor blockade or therapeutic mechanisms beyond acute 5-HT_{2A} agonism.

Safety Considerations and Clinical Vigilance

The divergent safety findings between the largest RCT showing serious adverse events including suicidal ideation and self-injury [1] and smaller studies showing no serious events [2-4, 7] likely reflect several factors. First, the larger sample (n=233) provided greater statistical power to detect rare serious events occurring at 1-2% base rates. Second, the 12-week follow-up period captured delayed-onset events beyond the acute treatment phase when most smaller studies ended follow-up. Third, the RCT's systematic safety monitoring may have identified events that went unreported or were attributed to underlying depression in less rigorous designs.

The finding that worsening suicidal state occurred across all dose groups including 1 mg control (9-17%) [1] suggests these events may represent the natural course of severe TRD rather than psilocybin-specific toxicity. However, the numerical trend toward higher rates in active treatment groups (14-17% vs 9%) and the occurrence of intentional self-injury and hospitalization indicate careful patient selection and monitoring protocols are essential. Studies excluding patients at significant suicide risk [1] demonstrated favorable safety profiles [2, 4, 7], suggesting this exclusion criterion should be maintained in clinical applications.

The universal occurrence of transient anxiety during drug onset [2, 4] and high rates of thought disorder or confusion (75%) [2] underscore the necessity of trained therapeutic support during dosing sessions. The lack of serious adverse events during supervised dosing sessions across all studies [1-4, 7] validates the safety of the therapeutic model when properly implemented.

Predictors and Mechanisms: Toward Personalized Treatment

The quality and content of the acute psychedelic experience consistently predicted treatment outcomes across studies [2–4, 6, 7], suggesting therapeutic effects depend partially on subjective phenomenology rather than pharmacology alone. This aligns with the importance of "set and setting" in psychedelic therapy and supports the integration of preparatory and integration psychotherapy [1–4, 6, 7] as essential treatment components rather than mere safety measures.

The identification of comorbid PTSD as a negative predictor [6] suggests trauma-related symptoms may interfere with psilocybin's antidepressant mechanisms or that depression secondary to PTSD represents a distinct syndrome requiring modified treatment approaches. This finding requires replication but has immediate clinical relevance for patient selection.

Neural mechanisms involving decreased network modularity [5], increased default mode network connectivity [8], and changes in executive and salience network flexibility [5] provide convergent evidence for psilocybin's effects on large-scale brain network organization. These findings support theoretical models proposing therapeutic benefit through enhanced neural flexibility and integration, potentially explaining both rapid onset (acute network perturbation) and gradual relapse (reversion to maladaptive patterns without sustained neuroplastic consolidation).

Population Considerations

The evidence base encompasses general TRD (≥ 2 failed treatments) [1, 2, 4] and severe TRD (≥ 5 failed treatments) [6], with both populations demonstrating treatment response. However, baseline characteristics varied substantially, with mean failed medications ranging from 2 to 4.6 [1, 2, 4, 5]. Most studies enrolled primarily unipolar depression [1, 2, 4–6, 8], with only one study explicitly including bipolar II disorder [9]. The generalizability of findings to bipolar depression, psychotic depression, and depression with active substance use disorders remains unknown due to systematic exclusion of these populations [1, 2, 9].

Most participants were White (78–92%) [1, 7] and many studies were conducted in single countries or limited geographic regions, raising questions about cross-cultural generalizability of psychedelic therapy paradigms that may depend on cultural context and expectancy effects.

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